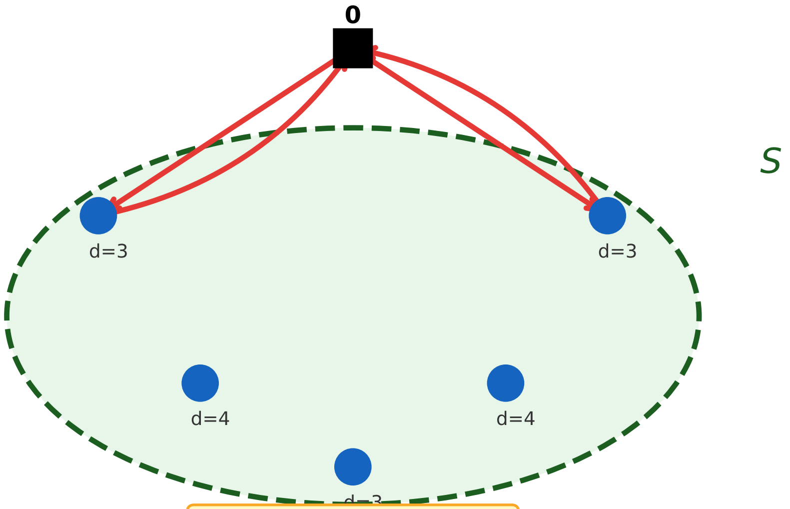
Cut edges $\delta(S)$ for subset $S$




$d(S) = \sum_{i \in S} d_i = 17,\ Q = 10$
RCI: $x(\delta(S)) \geq 2\lceil 17/10 \rceil = 4$

Rounded Capacity Inequality (RCI)

For every subset $S \subseteq V \setminus \{0\}$:

$$x(\delta(S)) \geq 2\left\lceil \frac{d(S)}{Q} \right\rceil$$

where $x(\delta(S))$ is the number of edges crossing the boundary of $S$, $d(S) = \sum_{i \in S} d_i$ is the total demand of $S$, and $Q$ is the vehicle capacity.

Intuition: if the total demand of $S$ exceeds $kQ$, then at least $k$ vehicles must enter (and leave) $S$, giving at least $2k$ crossing edges.